Now I will share some problem statements of dbms mysql ..give me simple code for this..in short which sufficient for create table and database..
Give me it in simple English language.
With also viva..questions with ans

Sure. Send the DBMS/MySQL problem statements one by one or all together.

For each problem statement, I'll give:

- Simple ER idea
- Short and easy MySQL code
- CREATE DATABASE + CREATE TABLE statements
- Simple INSERT values (if needed)
- Important SQL queries
- Viva questions with short answers
- Explanation in simple English

I'll keep the code:

- Easy to write in practical exam
- Short and sufficient
- Beginner-friendly
- Using simple SQL syntax only

Send the problem statements.

Create table Student (Sid, Name, Marks) table. Write and execute following PL/SQL statements:
i. Write a PL/SQL procedure to assign grade.
ii. Write a cursor-based PL/SQL block to display grades.
iii. Count students in each grade.

Student Table PL/SQL Program

1) Create Database and Table

```sql
CREATE DATABASE StudentDB;

USE StudentDB;

CREATE TABLE Student(
    Sid INT PRIMARY KEY,
    Name VARCHAR(30),
    Marks INT,
```

SQL

2) Insert Sample Data

```sql
INSERT INTO Student VALUES(1,'Amit',85,NULL);
INSERT INTO Student VALUES(2,'Neha',72,NULL);
```

SQL

i) PL/SQL Procedure to Assign Grade

Logic

- Marks >= 75 → A
- Marks >= 60 → B
- Marks >= 50 → C
- Else → D

```sql
DELIMITER //

CREATE PROCEDURE AssignGrade()
BEGIN

UPDATE Student
SET Grade =
CASE
    WHEN Marks >= 75 THEN 'A'
    WHEN Marks >= 60 THEN 'B'
    WHEN Marks >= 50 THEN 'C'
    ELSE 'D'
END;
```

END //

SQL

Execute Procedure

SQL

ii) Cursor-Based Block to Display Grades

```sql
DELIMITER //

CREATE PROCEDURE DisplayGrades()
BEGIN

DECLARE done INT DEFAULT 0;
DECLARE sname VARCHAR(30);
DECLARE g CHAR(1);

DECLARE cur CURSOR FOR
SELECT Name, Grade FROM Student;

DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

OPEN cur;

read_loop: LOOP

FETCH cur INTO sname, g;

IF done = 1 THEN
    LEAVE read_loop;
END IF;

SELECT sname AS Name, g AS Grade;

END LOOP;

CLOSE cur;

END //
```

SQL

Execute

SQL

iii) Count Students in Each Grade

```sql
SELECT Grade, COUNT(*) AS Total_Students
```

SQL

Output Table Example

Sid	Name	Marks	Grade
1	Amit	85	A
2	Neha	72	B
3	Rahul	60	B
4	Pooja	45	D

Viva Questions with Answers

1) What is PL/SQL?

PL/SQL is Procedural Language used with SQL for programming.

2) What is a Procedure?

Procedure is a stored program that performs a task.

3) What is Cursor?

Cursor is used to fetch records one by one from a table.

4) Why PRIMARY KEY is used?

PRIMARY KEY uniquely identifies each row.

5) What is CASE statement?

CASE is used for conditional checking.

6) What is GROUP BY?

GROUP BY is used to group similar data.

7) Difference between SQL and PL/SQL?

SQL	PL/SQL
Query language	Programming language
Executes single query	Executes block of code

Important Short Commands for Viva

```
SHOW DATABASES;
SHOW TABLES;
```

SQL

Create Employee (Eid, Ename, Salary) table. Write and execute following PL/SQL statements:
i. Write a PL/SQL function to classify salary. ii. Display classification (Class I, Class II, etc).
iii. Count employees in each class.

Employee Table PL/SQL Program

1) Create Database and Table

```sql
CREATE DATABASE EmployeeDB;

USE EmployeeDB;

CREATE TABLE Employee(
    Eid INT PRIMARY KEY,
    Ename VARCHAR(30),
    Salary INT,
```

SQL

2) Insert Sample Data

```sql
INSERT INTO Employee VALUES(1,'Amit',60000,NULL);
INSERT INTO Employee VALUES(2,'Neha',45000,NULL);
```

SQL

i) PL/SQL Function to Classify Salary

Logic

- Salary >= 50000 → Class I
- Salary >= 40000 → Class II
- Salary >= 25000 → Class III
- Else → Class IV

```sql
DELIMITER //

CREATE FUNCTION GetClass(sal INT)
RETURNS VARCHAR(20)

DETERMINISTIC

BEGIN

DECLARE emp_class VARCHAR(20);

IF sal >= 50000 THEN
    SET emp_class = 'Class I';
```

```
ELSEIF sal >= 40000 THEN
    SET emp_class = 'Class II';

ELSEIF sal >= 25000 THEN
    SET emp_class = 'Class III';

ELSE
    SET emp_class = 'Class IV';

END IF;

RETURN emp_class;

END //
```

SQL

ii) Display Classification

Update Employee Class

SQL

Display Table

SQL

iii) Count Employees in Each Class

```
SELECT Class, COUNT(*) AS Total_Employees
```

SQL

Output Example

Eid	Ename	Salary	Class
1	Amit	60000	Class I
2	Neha	45000	Class II
3	Rahul	30000	Class III
4	Pooja	18000	Class IV

Viva Questions with Answers

1) What is Function in PL/SQL?

Function is a stored program that returns a value.

2) Difference between Function and Procedure?

Function	Procedure
Returns value	Does not return value
Used in query	Called separately

3) What is DELIMITER?

DELIMITER changes the end symbol of SQL statement.

4) What is RETURN in Function?

RETURN sends value back from function.

5) What is GROUP BY?

GROUP BY groups similar rows together.

6) What is DETERMINISTIC?

It means function gives same output for same input.

7) Why PRIMARY KEY is used?

To uniquely identify each employee.

Important Commands

```sql
SHOW DATABASES;

SHOW TABLES;

DESC Employee;
```

SQL

Create Product (Pid, Pname, Price) table. Write and execute following PL/SQL statements:
i. Write a PL/SQL procedure to calculate discount. ii. Display updated prices.
iii. Find maximum priced product.

Product Table PL/SQL Program

1) Create Database and Table

```sql
CREATE DATABASE ProductDB;

USE ProductDB;

CREATE TABLE Product(
    Pid INT PRIMARY KEY,
    Pname VARCHAR(30),
```

SQL

2) Insert Sample Data

```sql
INSERT INTO Product VALUES(1,'Laptop',50000);
INSERT INTO Product VALUES(2,'Mobile',20000);
```

SQL

i) PL/SQL Procedure to Calculate Discount

Logic

- 10% discount on all products

```sql
DELIMITER //

CREATE PROCEDURE DiscountPrice()
BEGIN

UPDATE Product
SET Price = Price - (Price * 0.10);

END //
```

SQL

Execute Procedure

SQL

ii) Display Updated Prices

SQL

iii) Find Maximum Priced Product

```sql
SELECT *
```

SQL

Output Example

Pid	Pname	Price
1	Laptop	45000
2	Mobile	18000
3	TV	27000
4	Mouse	900

Viva Questions with Answers

1) What is Procedure?

Procedure is a stored SQL program used to perform task.

2) What is UPDATE statement?

UPDATE changes existing data in table.

3) What is MAX() function?

MAX() returns highest value from column.

4) What is PRIMARY KEY?

PRIMARY KEY uniquely identifies records.

5) Why DELIMITER is used?

DELIMITER changes statement ending symbol.

6) What is Aggregate Function?

Functions like MAX(), MIN(), COUNT(), AVG() are aggregate functions.

7) Difference between DELETE and DROP?

DELETE	DROP
Deletes rows	Deletes whole table
Table exists	Table removed

Important Commands

```sql
SHOW DATABASES;

SHOW TABLES;

DESC Product;
```

SQL

Create Issue (Bid, Mid, IssueDate, ReturnDate) table.Write and execute following PL/SQL statements:
i. Write a PL/SQL function to calculate fine. ii. Display fine.
iii. Find highest fine.

Issue Table PL/SQL Program

1) Create Database and Table

```sql
CREATE DATABASE LibraryDB;

USE LibraryDB;
```

```sql
CREATE TABLE Issue(
    Bid INT,
    Mid INT,
    IssueDate DATE,
    ReturnDate DATE,
```

SQL

2) Insert Sample Data

```sql
INSERT INTO Issue VALUES
(1,101,'2025-01-01','2025-01-10',NULL);

INSERT INTO Issue VALUES
(2,102,'2025-01-01','2025-01-20',NULL);
```

SQL

i) PL/SQL Function to Calculate Fine

Logic

- Fine = Rs.10 per extra day after 7 days

```sql
DELIMITER //

CREATE FUNCTION CalculateFine(
    issue_date DATE,
    return_date DATE
)

RETURNS INT

DETERMINISTIC

BEGIN

DECLARE days INT;
DECLARE fine_amt INT;

SET days = DATEDIFF(return_date, issue_date);

IF days > 7 THEN
```

```
    SET fine_amt = (days - 7) * 10;
ELSE
    SET fine_amt = 0;
END IF;

RETURN fine_amt;
```

SQL

Update Fine in Table

SQL

ii) Display Fine

SQL

iii) Find Highest Fine

```
SELECT *
```

SQL

Output Example

Bid	Mid	IssueDate	ReturnDate	Fine
1	101	2025-01-01	2025-01-10	20
2	102	2025-01-01	2025-01-20	120

Bid	Mid	IssueDate	ReturnDate	Fine
3	103	2025-01-01	2025-01-05	0

Viva Questions with Answers

1) What is Function in PL/SQL?

Function is a stored program that returns value.

2) What is DATEDIFF()?

DATEDIFF() finds difference between two dates.

3) What is MAX() function?

MAX() returns highest value.

4) What is UPDATE statement?

UPDATE changes existing records.

5) What is DETERMINISTIC?

Same input gives same output.

6) Why Fine column is used?

To store calculated fine amount.

7) Difference between DATE and INT datatype?

DATE	INT
Stores date	Stores numbers

Important Commands

```sql
SHOW DATABASES;

SHOW TABLES;

DESC Issue;
```

SQL

Create Account (Aid, Balance) table. Write and execute following statements:
i. Create a trigger to prevent negative balance. ii. Verify using update statements.
iii. Display minimum balance.

Account Table Trigger Program

1) Create Database and Table

```sql
CREATE DATABASE BankDB;

USE BankDB;

CREATE TABLE Account(
    Aid INT PRIMARY KEY,
```

SQL

2) Insert Sample Data

```sql
INSERT INTO Account VALUES(1,5000);
```

SQL

i) Create Trigger to Prevent Negative Balance

```sql
DELIMITER //

CREATE TRIGGER CheckBalance
BEFORE UPDATE
ON Account
FOR EACH ROW

BEGIN

IF NEW.Balance < 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Negative balance not allowed';
END IF;

END //
```

SQL

ii) Verify Using UPDATE Statements

Valid Update

```sql
UPDATE Account
```

SQL

Invalid Update

```sql
UPDATE Account
```

SQL

Output

```
ERROR: Negative balance not allowed
```

Display Table

SQL

iii) Display Minimum Balance

SQL

Output Example

Aid	Balance
1	2000
2	8000
3	3000

Viva Questions with Answers

1) What is Trigger?

Trigger is automatic action performed when event occurs.

2) What is BEFORE UPDATE Trigger?

It runs before updating records.

3) What is NEW keyword?

NEW refers to new value being updated.

4) What is SIGNAL statement?

SIGNAL displays custom error message.

5) What is MIN() function?

MIN() returns smallest value.

6) Why PRIMARY KEY is used?

To uniquely identify account.

7) Difference between Trigger and Procedure?

Trigger	Procedure
Automatic execution	Manual execution
Event based	User called

Important Commands

```sql
SHOW DATABASES;

SHOW TABLES;

DESC Account;
```

SQL

Create Employee (Eid, Ename, Salary) table. Write and execute following statements:
i. Create a trigger to enforce salary range (20000–80000).
ii. Update and verify.
iii. Display average salary.

Employee Table Trigger Program

1) Create Database and Table

```sql
CREATE DATABASE CompanyDB;

USE CompanyDB;

CREATE TABLE Employee(
    Eid INT PRIMARY KEY,
    Ename VARCHAR(30),
```

SQL

2) Insert Sample Data

```sql
INSERT INTO Employee VALUES(1,'Amit',30000);
```

SQL

i) Create Trigger to Enforce Salary Range

Condition

- Salary must be between 20000 and 80000

```sql
DELIMITER //

CREATE TRIGGER SalaryCheck
BEFORE UPDATE
ON Employee
FOR EACH ROW

BEGIN

IF NEW.Salary < 20000 OR NEW.Salary > 80000 THEN

SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT = 'Salary must be between 20000 and 80000';

END IF;

END //
```

SQL

```
DELIMITER ;
```

ii) Update and Verify

Valid Update

```
UPDATE Employee
```

SQL

Invalid Update

```
UPDATE Employee
```

SQL

Output

```
ERROR: Salary must be between 20000 and 80000
```

Display Table

SQL

iii) Display Average Salary

SQL

Output Example

Eid	Ename	Salary
1	Amit	60000
2	Neha	50000
3	Rahul	70000

Viva Questions with Answers

1) What is Trigger?

Trigger is automatic action performed on database event.

2) What is BEFORE UPDATE Trigger?

It executes before update operation.

3) What is NEW keyword?

NEW represents new updated value.

4) What is AVG() function?

AVG() calculates average value.

5) Why SIGNAL is used?

SIGNAL displays custom error message.

6) What is PRIMARY KEY?

PRIMARY KEY uniquely identifies each employee.

7) Difference between AVG() and SUM()?

AVG()	SUM()
Finds average	Finds total

Important Commands

```sql
SHOW DATABASES;

SHOW TABLES;

DESC Employee;
```

SQL

Create Product(Pid) and Orders (Oid, Pid) tables.Write and execute following statements:
i. Create a trigger for delete cascade. ii. Verify deletion.
iii. Count remaining orders.

Product and Orders Trigger Program

1) Create Database and Tables

```sql
CREATE DATABASE ShopDB;

USE ShopDB;

CREATE TABLE Product(
    Pid INT PRIMARY KEY
);

CREATE TABLE Orders(
    Oid INT PRIMARY KEY,
```

SQL

2) Insert Sample Data

```sql
INSERT INTO Product VALUES(1);
INSERT INTO Product VALUES(2);

INSERT INTO Orders VALUES(101,1);
```

SQL

i) Create Trigger for Delete Cascade

When product is deleted, related orders also deleted

```sql
DELIMITER //

CREATE TRIGGER DeleteCascade
BEFORE DELETE
ON Product
FOR EACH ROW

BEGIN

DELETE FROM Orders
WHERE Pid = OLD.Pid;

END //
```

SQL

ii) Verify Deletion

Delete Product

SQL

Display Product Table

SQL

SQL

Display Orders Table

SQL

iii) Count Remaining Orders

SQL

Output Example

Product Table

Pid
2

Orders Table

Oid	Pid
103	2

Viva Questions with Answers

1) What is Trigger?

Trigger is automatic SQL action.

2) What is DELETE Trigger?

It runs automatically during delete operation.

3) What is OLD keyword?

OLD stores old record values before deletion.

4) What is Cascade Delete?

Deleting parent record also deletes child records.

5) What is COUNT(*)?

COUNT(*) counts total rows.

6) Why PRIMARY KEY is used?

To uniquely identify records.

7) Difference between BEFORE DELETE and AFTER DELETE?

BEFORE DELETE	AFTER DELETE
Runs before deletion	Runs after deletion

Important Commands

```sql
SHOW DATABASES;

SHOW TABLES;

DESC Product;

DESC Orders;

SELECT * FROM Product;
```

SQL

Create Student (Sid, Marks) table. Write and execute following statements:
i. Create a trigger to validate marks (0–100). ii. Update and verify.
iii. Display maximum marks.

Student Table Trigger Program

1) Create Database and Table

```
CREATE DATABASE SchoolDB;

USE SchoolDB;

CREATE TABLE Student(
    Sid INT PRIMARY KEY,
```

SQL

2) Insert Sample Data

```
INSERT INTO Student VALUES(1,85);
```

SQL

i) Create Trigger to Validate Marks

Condition

- Marks must be between 0 and 100

```
DELIMITER //

CREATE TRIGGER CheckMarks
BEFORE UPDATE
ON Student
```

```sql
FOR EACH ROW

BEGIN

IF NEW.Marks < 0 OR NEW.Marks > 100 THEN

SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT = 'Marks must be between 0 and 100';

END IF;

END //
```

SQL

ii) Update and Verify

Valid Update

```sql
UPDATE Student
```

SQL

Invalid Update

```sql
UPDATE Student
```

SQL

Output

```
ERROR: Marks must be between 0 and 100
```

Display Table

SQL

iii) Display Maximum Marks

SQL

Output Example

Sid	Marks
1	90
2	72
3	60

1) What is Trigger?

Trigger is automatic action executed on database event.

2) What is BEFORE UPDATE Trigger?

It runs before updating records.

3) What is NEW keyword?

NEW represents new value after update.

4) What is MAX() function?

MAX() returns highest value.

5) Why SIGNAL statement is used?

To display custom error message.

6) What is PRIMARY KEY?

PRIMARY KEY uniquely identifies records.

7) Difference between MAX() and MIN()?

MAX()	MIN()
Highest value	Lowest value

Important Commands

SQL

```sql
SHOW DATABASES;

SHOW TABLES;

DESC Student;

SELECT * FROM Student;
```